

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_failure_replay.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay.
- Best synthetic holdout gap: atsp_no_replay.

Run Metadata

- run_name: run_20260513_183147_q
- started_at_local: 2026-05-13 18:31:47
- finished_at_local: 2026-05-13 18:40:32
- duration_hhmm: 00:09
- duration_seconds: 525.155
- seed_offset: 16000
- replicate_label: q
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.183339	0.009004	0.096172	0.901717	0.58	0.019985
atsp_random_replay	random	score_only	False	0.188164	0.030781	0.109473	0.0	0.78	0.0
atsp_failure_replay	failure	score_only	False	0.151975	0.013889	0.082932	0.711538	0.78	0.081453
atsp_failure_replay_compression	failure	novelty_gate	True	0.193498	0.009004	0.101251	0.858717	0.58	0.036301

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.183339, synthetic 0.009004, combined 0.096172.

- Accepted-epoch count 3, mean accepted code novelty 0.901717, and final complexity 0.58.
- Adaptation efficiency 0.019985 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.183339 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.003915, ry=0.071419.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.188164, synthetic 0.030781, combined 0.109473.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.188164 across 4 instances; family means: ft=0.32281, ftv=0.269684, p=0.014413, ry=0.14575.
- Panel synthetic_holdout mean gap 0.030781 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.037511, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1649, "family": "ftv", "name": "ftv33", "optimality_gap": 0.282271}, {"best_known_cost": 1530, "cost": 1943, "family": "ftv", "name": "ftv38", "optimality_gap": 0.269935}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.151975, synthetic 0.013889, combined 0.082932.
- Accepted-epoch count 2, mean accepted code novelty 0.711538, and final complexity 0.78.
- Adaptation efficiency 0.081453 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.151975 across 4 instances; family means: ft=0.275018, ftv=0.233726, p=0.002847, ry=0.096311.
- Panel synthetic_holdout mean gap 0.013889 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.048593, wind_clusters=0.006965.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1551, "family": "ftv", "name": "ftv33", "optimality_gap": 0.206065}, {"best_known_cost": 1530, "cost": 1779, "family": "ftv", "name": "ftv38", "optimality_gap": 0.162745}, {"best_known_cost": 1473, "cost": 1710, "family": "ftv", "name": "ftv35", "optimality_gap": 0.160896}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.193498, synthetic 0.009004, combined 0.101251.
- Accepted-epoch count 2, mean accepted code novelty 0.858717, and final complexity 0.58.

- Adaptation efficiency 0.036301 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193498 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004093, ry=0.112744.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

Summary of Replay-Aware ATSP Benchmark Conditions

Condition	Replay Mode	TSP LIB Gap	Synthetic Gap	Transfer Gap	Code Novelty (mean)	Complexity	Notes on Transfer & Novelty
atsp_no_replay	None	0.183339	**0.009004**	0.096172	0.9017	0.58	Best synthetic gap; moderate transfer gap; highest code novelty with no replay.
atsp_random_replay	Random	0.188164	0.030781	0.109473	0.0	0.78	Worst transfer and synthetic gaps; zero code novelty.
atsp_failure_replay	Failure replay	**0.151975**	0.013889	**0.082932**	0.7115	0.78	Best TSPLIB and transfer gaps; lower code novelty than no replay but transfer improved.
atsp_failure_replay_compression	Failure replay + compression	0.193498	0.009004	0.101251	0.8587	0.58	Compression reduces complexity and novelty; higher TSPLIB gap than failure replay without compression, transfer gap also higher.

Key Interpretations

- ****Main Transfer Evidence (TSPLIB & synthetic holdout gaps):****
- The best ****transfer**** condition is ****atsp_failure_replay**** with the lowest TSPLIB gap (0.151975) and lowest transfer gap (0.082932).
- ****atsp_no_replay**** performed best on synthetic holdout data (0.009004 synthetic gap), but TSPLIB gap is higher (0.183339), indicating less transfer robustness.
- ****Code Novelty vs. Transfer:****
- ****No replay**** (atsp_no_replay) has the highest code novelty (~0.9) but worse transfer metrics, indicating high lexical novelty without corresponding algorithmic improvements as measured.
- ****Failure replay**** conditions reduce code novelty (~0.71-0.86) but improve transfer, showing a trade-off: transfer improves even as code novelty drops.
- ****Random replay**** yields no code novelty and worse transfer, confirming poor adaptation without targeted replay.

- **Replay Modes and Effects:**
- **No replay:** Highest code novelty but suboptimal transfer and TSPLIB gaps.
- **Random replay:** Zero novelty and worst transfer performance, suggesting ineffective replay strategy.
- **Failure replay:** Best transfer performance, confirming replay focusing on failures aids generalization and transfer.
- **Failure replay with compression:** Complexity and code novelty reduced relative to failure replay alone, but transfer and TSPLIB gaps increase slightly, implying compression pressure trades off transfer optimality.

Conservative Conclusions

- Failure replay significantly improves benchmark transfer (TSPLIB and synthetic holdout gaps) compared to no replay or random replay, despite lower code novelty.
- High code novelty alone (seen in no replay) does not guarantee better transfer, so lexical novelty should not be conflated with algorithmic invention.
- Random replay and compression-aware replay conditions result in inferior or comparable transfer relative to pure failure replay, indicating the importance of replay type and overhead.
- Compression-aware replay reduces complexity and novelty but at a small cost to transfer metrics, suggesting a trade-off between compactness and transfer optimality.

Recommendations for Use

- Prioritize **atsp_failure_replay** for best overall transfer in replay-aware ATSP heuristics.
- Avoid random replay due to lack of novelty and poor transfer.
- Consider cost-benefit of compression-aware failure replay if lower complexity is required but note possible transfer loss.
- Code novelty metrics should be interpreted cautiously and not as proof of algorithmic advancement without corroborating transfer improvements.